

Using UDP_Sniffer to monitor data being broadcast to a network

With the issues we have been having with SCS data not passing between the mission and science networks, it can be helpful to verify that a computer can in fact see the SCS broadcast data. I wrote a simple command line python script that will open a collection of UDP ports, listen for data, and when data is received, print out some information about the sender, the port the data was received on, and the data itself (if the data can be converted to ASCII.)

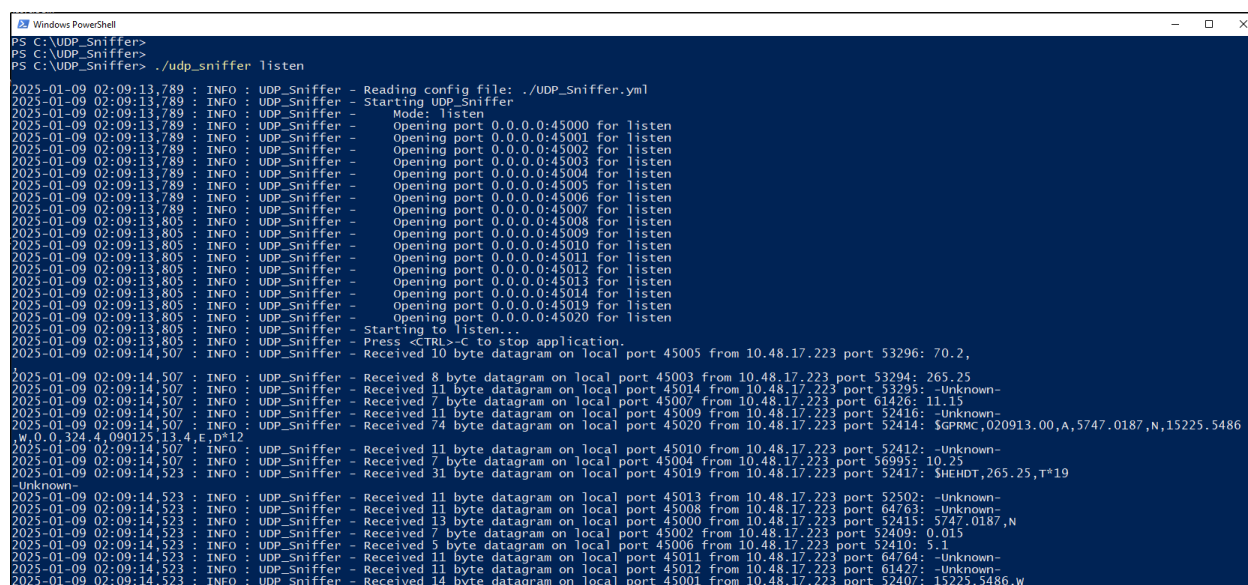
This version of instructions is for use with the compiled version of UDP_Sniffer >= 1.1. The compiled version will run without an installation of Python. Versions >= 1.1 can run in both “listen” mode and “tx” mode. “listen” mode will open the ports for listening while “tx” mode will transmit data and can be used for testing when an SCS server is not available. This document will only cover using UDP_Sniffer in listen mode.

To use UDP_Sniffer to monitor UDP ports for SCS broadcast data, open a command prompt or powershell and navigate to the UDP_Sniffer folder.

Start the UDP_Sniffer program specifying “listen” mode:

```
C:\UDP_Sniffer>UDP_Sniffer listen
```

The output will look something like the image below. UDP_Sniffer will first report the IP address and ports that it is opening to listen on. Note that an IP address of 0.0.0.0 means all interfaces. In winter 2024 we worked with OMAO IT to open up ports 45000-45020 for SCS data and UDP_Sniffer is configured to listen on these ports.



```
Windows PowerShell
PS C:\UDP_Sniffer>
PS C:\UDP_Sniffer>
PS C:\UDP_Sniffer> ./udp_sniffer listen
2025-01-09 02:09:13,789 INFO : UDP_Sniffer - Reading config file: ./UDP_Sniffer.yml
2025-01-09 02:09:13,789 INFO : UDP_Sniffer - Starting UDP_Sniffer
2025-01-09 02:09:13,789 INFO : UDP_Sniffer - Mode: listen
2025-01-09 02:09:13,789 INFO : UDP_Sniffer - Opening port 0.0.0.0:45000 for listen
2025-01-09 02:09:13,789 INFO : UDP_Sniffer - Opening port 0.0.0.0:45001 for listen
2025-01-09 02:09:13,789 INFO : UDP_Sniffer - Opening port 0.0.0.0:45002 for listen
2025-01-09 02:09:13,789 INFO : UDP_Sniffer - Opening port 0.0.0.0:45003 for listen
2025-01-09 02:09:13,789 INFO : UDP_Sniffer - Opening port 0.0.0.0:45004 for listen
2025-01-09 02:09:13,789 INFO : UDP_Sniffer - Opening port 0.0.0.0:45005 for listen
2025-01-09 02:09:13,789 INFO : UDP_Sniffer - Opening port 0.0.0.0:45006 for listen
2025-01-09 02:09:13,789 INFO : UDP_Sniffer - Opening port 0.0.0.0:45007 for listen
2025-01-09 02:09:13,805 INFO : UDP_Sniffer - Opening port 0.0.0.0:45008 for listen
2025-01-09 02:09:13,805 INFO : UDP_Sniffer - Opening port 0.0.0.0:45009 for listen
2025-01-09 02:09:13,805 INFO : UDP_Sniffer - Opening port 0.0.0.0:45010 for listen
2025-01-09 02:09:13,805 INFO : UDP_Sniffer - Opening port 0.0.0.0:45011 for listen
2025-01-09 02:09:13,805 INFO : UDP_Sniffer - Opening port 0.0.0.0:45012 for listen
2025-01-09 02:09:13,805 INFO : UDP_Sniffer - Opening port 0.0.0.0:45013 for listen
2025-01-09 02:09:13,805 INFO : UDP_Sniffer - Opening port 0.0.0.0:45014 for listen
2025-01-09 02:09:13,805 INFO : UDP_Sniffer - Opening port 0.0.0.0:45019 for listen
2025-01-09 02:09:13,805 INFO : UDP_Sniffer - opening port 0.0.0.0:45020 for listen
2025-01-09 02:09:13,805 INFO : UDP_Sniffer - Starting to listen...
2025-01-09 02:09:14,507 INFO : UDP_Sniffer - Press <CTRL-C> to stop application.
2025-01-09 02:09:14,507 INFO : UDP_Sniffer - Received 10 byte datagram on local port 45005 from 10.48.17.223 port 53296: 70.2,
2025-01-09 02:09:14,507 INFO : UDP_Sniffer - Received 8 byte datagram on local port 45003 from 10.48.17.223 port 53294: 265.25
2025-01-09 02:09:14,507 INFO : UDP_Sniffer - Received 11 byte datagram on local port 45014 from 10.48.17.223 port 53295: -unknown-
2025-01-09 02:09:14,507 INFO : UDP_Sniffer - Received 7 byte datagram on local port 45007 from 10.48.17.223 port 61426: 11.15
2025-01-09 02:09:14,507 INFO : UDP_Sniffer - Received 11 byte datagram on local port 45009 from 10.48.17.223 port 52416: -unknown-
2025-01-09 02:09:14,507 INFO : UDP_Sniffer - Received 74 byte datagram on local port 45020 from 10.48.17.223 port 52414: SGPBMC,020913.00,A,5747.0187,N,15225.5486
w 0.0.324.4,090125.13.4 E.D*12
2025-01-09 02:09:14,507 INFO : UDP_Sniffer - Received 11 byte datagram on local port 45010 from 10.48.17.223 port 52412: -unknown-
2025-01-09 02:09:14,507 INFO : UDP_Sniffer - Received 7 byte datagram on local port 45004 from 10.48.17.223 port 56995: 10.25
2025-01-09 02:09:14,523 INFO : UDP_Sniffer - Received 31 byte datagram on local port 45019 from 10.48.17.223 port 52417: $HEHDT,265.25,T*19
Unknown-
2025-01-09 02:09:14,523 INFO : UDP_Sniffer - Received 11 byte datagram on local port 45013 from 10.48.17.223 port 52502: -unknown-
2025-01-09 02:09:14,523 INFO : UDP_Sniffer - Received 11 byte datagram on local port 45008 from 10.48.17.223 port 64763: -unknown-
2025-01-09 02:09:14,523 INFO : UDP_Sniffer - Received 13 byte datagram on local port 45000 from 10.48.17.223 port 52415: 5747.0187,N
2025-01-09 02:09:14,523 INFO : UDP_Sniffer - Received 7 byte datagram on local port 45002 from 10.48.17.223 port 52409: 0.015
2025-01-09 02:09:14,523 INFO : UDP_Sniffer - Received 5 byte datagram on local port 45006 from 10.48.17.223 port 52410: 5.1
2025-01-09 02:09:14,523 INFO : UDP_Sniffer - Received 11 byte datagram on local port 45011 from 10.48.17.223 port 64764: -unknown-
2025-01-09 02:09:14,523 INFO : UDP_Sniffer - Received 11 byte datagram on local port 45012 from 10.48.17.223 port 61427: -unknown-
2025-01-09 02:09:14,523 INFO : UDP_Sniffer - Received 14 byte datagram on local port 45001 from 10.48.17.223 port 52407: 15225.5486,w
```

When data is received, information about it including the source IP address, port, number of

bytes, and the data itself (if it can be converted to ASCII) will be printed to the screen. You should expect to see data from all ports that are currently configured in SCS to send data. The list of sensor data and port numbers can be found [here](#).

In order for UDP_Sniffer to work, you must configure a Windows Firewall exception for it. You can either create a new inbound UDP port rule for the range of ports 45000-45020 or you can create an exception by clicking “Allow and app or feature.” Note that adding an inbound port rule will be more flexible than “allowing the app” as app based exceptions are specific to an executable and if you move the executable or try to run a different version from a different folder, it will no longer work.

While out of the scope of this document, be aware that the configuration for UDP_Sniffer is contained in the UDP_Sniffer.yml file. It is currently configured to listen to the Dyson SCS UDP ports and should not need to be edited.